

Julretsu

User Manual

The spreadsheet that catches your mistakes before they cost you.

Version 1.2.0

Circuitspecter Studio

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1. Welcome to Julretsu

Julretsu is a desktop spreadsheet for Windows, macOS and Linux. It works like the spreadsheets you already know: a grid of cells, formulas that recalculate as you type, formatting, sorting, charts and printing. Your workbooks stay on your own computer.

What makes Julretsu different is its **safety net**. Spreadsheet mistakes such as a mistyped amount, a total that skips a row or a paste that lands one row off cost businesses real money every year. Julretsu watches for these as you work, explains what looks wrong in plain language, offers one click fixes, and keeps a history of every change so you can always see who changed what and when.

What you can do with Julretsu

- Build budgets, price lists, schedules and reports on sheets of up to 1,000,000 rows and 16,384 columns.
- Use formulas such as `SUM`, `AVERAGE` and `IF`, and link one worksheet to another.
- Format cells with fonts, colours, borders, currency, percentages and dates.
- Open and save CSV files and Microsoft Excel `.xlsx` workbooks.
- Turn a selection into a bar or line chart, export it as a report, print it, or save it as a PDF.
- Work in English, Korean or Japanese, and type any of them into your cells.
- Automate repetitive edits with small Lua macros.
- Optionally ask an AI model to propose edits, which you review before anything changes.

Tip. Julretsu opens with an example project budget. Every screenshot in this manual uses that example, so you can follow along.

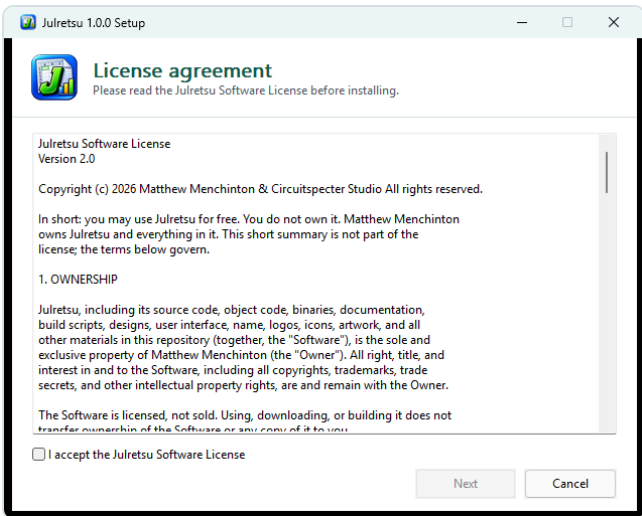
2. Installing Julretsu

Download the installer for your computer from the Julretsu website. Julretsu 1.2.0 comes in these packages:

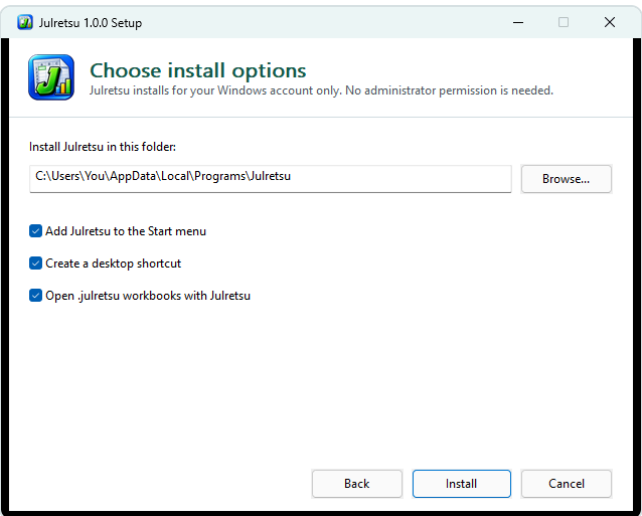
Computer	File	Requirements
Windows	Julretsu-1.2.0-Windows-Setup.exe	Windows 10 or 11 (64 bit), graphics that support OpenGL 3.3
Mac	Julretsu-1.2.0-macOS.dmg	macOS 13.3 Ventura or later, Apple silicon or Intel
Linux (Ubuntu, Debian, Mint)	Julretsu-1.2.0-linux-amd64.deb	64 bit Intel or AMD, a desktop with X11 or XWayland
Linux (any distribution)	Julretsu-1.2.0-linux-x86_64.tar.gz	Same as above; no installation needed

Windows

1. Double-click **Julretsu-1.2.0-Windows-Setup.exe**.
2. Read the license, tick **I accept the Julretsu Software License**, and click **Next**.
3. Choose where to install Julretsu. The suggested folder is inside your own user folder, so you do not need administrator permission.
4. Choose your options: a Start menu entry, a desktop shortcut, and whether Julretsu should open `.julretsu` workbooks when you double-click them.
5. Click **Install**. When it finishes, leave **Start Julretsu now** ticked and click **Finish**.



Step 2: the license agreement



Steps 3 and 4: install options

If Windows shows "Windows protected your PC". This message appears for new programs that Windows has not seen many times yet. Click **More info**, check that the publisher and file name are what you expect, then click **Run anyway**.

Mac

1. Double-click **Julretsu-1.2.0-macOS.dmg** to open it.
2. Drag **Julretsu** onto the **Applications** folder shown in the same window.
3. Open your Applications folder and start Julretsu.

If macOS says it cannot check the app for malicious software. The first time, Control-click (or right-click) Julretsu in Applications and choose **Open**, then click **Open** again. You can also go to **System Settings, Privacy & Security** and click **Open Anyway**. You only need to do this once.

Linux

Ubuntu, Debian, Linux Mint and similar: double-click the `.deb` file to install it with your software centre, or run this command in the folder where you downloaded it:

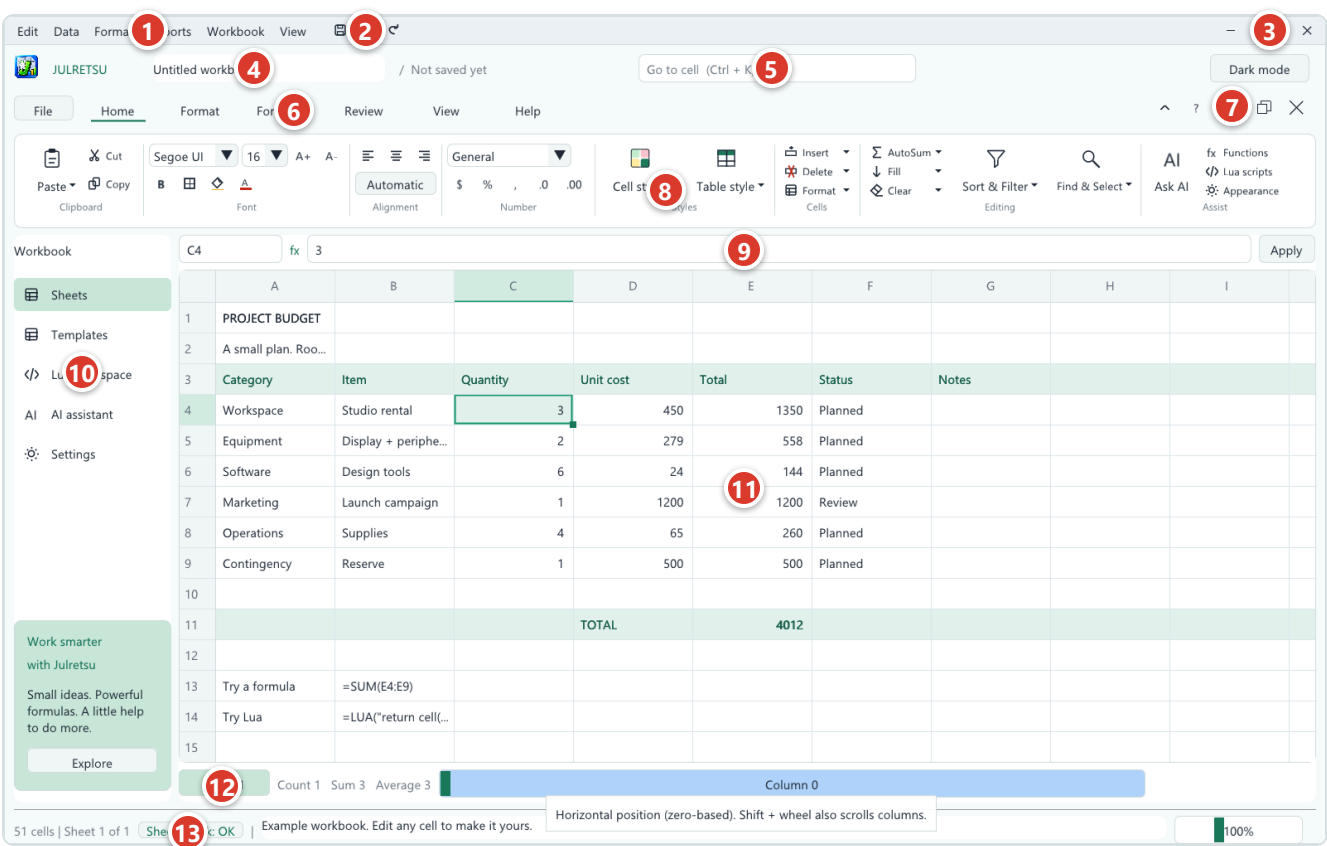
```
sudo apt install ./Julretsu-1.2.0-linux-amd64.deb
```

Julretsu then appears in your applications menu, and `.julretsu` files open in Julretsu.

Other distributions: extract the `.tar.gz` file anywhere you like and run `bin/julretsu` inside the extracted folder.

Note. On Linux, Julretsu uses the desktop's own file dialogs. If you see a message asking for zenity or kdialog, install one of them, for example `sudo apt install zenity`.

3. A tour of the window



The main window with the example budget

#	Part	What it does
1	Menus	Edit, Data, Format, Reports, Workbook and View menus with every command.
2	Quick access	Save, Undo and Redo, always one click away.
3	Window buttons	Minimize, maximize and close Julretsu. Drag any empty part of the top two rows to move the window.
4	Workbook name	The file you are working on and whether it has unsaved changes.
5	Go to cell	Type an address such as C4 and press Enter to jump there (Ctrl + K).
6	File and ribbon tabs	File opens the file menu. The tabs switch the ribbon: Home, Format, Formulas, Review, View and Help.
7	Workbook buttons	Collapse the ribbon, open help, and minimize, restore or close the workbook.
8	Ribbon	The most used commands for the selected tab, grouped like other office programs.
9	Formula bar	Shows the address and the exact contents of the selected cell. Edit here and press Enter or click Apply .
10	Navigation	Sheets, templates, the Lua workspace, the AI assistant and settings.
11	Grid	Your cells. The active cell has a green outline and a small fill handle in its corner.
12	Sheet button and totals	Switch worksheets. The count, sum and average of the selected cells appear beside it.
13	Status bar	Number of filled cells, the Sheet check badge, messages, and the zoom slider.

4. Entering and editing data

Selecting cells

- Click a cell to select it. Use the arrow keys, `Tab`, `Page Up` and `Page Down` to move.
- `Shift` +click another cell to select the rectangle between them.
- Click a row number to select the whole row. `Ctrl` +click adds or removes rows, and `Shift` +click selects a run of rows.
- `Ctrl` + `Home` jumps to A1. Edit, then **Select used range** selects everything that holds data.

Typing and changing values

1. Double-click a cell or press `F2`, or click in the formula bar.
2. Type a number, text, `TRUE` or `FALSE`, or a formula that starts with `=`.
3. Press `Enter` to keep the change, or `Esc` to cancel it.

Tip. To store text that starts with `=` or looks like a number, start it with an apostrophe. For example `'=hello` or `'00123` are kept exactly as typed.

Long text stays inside its own cell and ends with "...". Hover over the cell, or look at the formula bar, to read all of it. A number that is too wide for its cell shows `###` so it can never be misread; zoom in, or read it in the formula bar.

Undo and redo

Press `Ctrl` + `Z` to undo and `Ctrl` + `Y` to redo, or use the quick access buttons. A paste, a sort or a macro is undone in one step.

Copy, cut and paste

- `Ctrl` + `C` copies, `Ctrl` + `X` cuts and `Ctrl` + `V` pastes. You can copy up to 10,000 cells at a time.
- `Ctrl` + `Shift` + `V` pastes values only, without formulas or formatting.
- Copying inside Julretsu keeps formulas and formatting. Relative references move with the paste, while references with `$` stay fixed.
- You can paste text copied from other programs, such as rows from a web page or another spreadsheet.

Filling a series

Drag the small square at the corner of your selection down or to the right. Formulas are copied with their references adjusted. If you start with two numbers, such as 5 and 10, Julretsu continues the series: 15, 20, 25 and so on. You can also use **Home, Editing, Fill** to fill down or right.

Finding things

Press **Ctrl** + **F** to search values and formulas on the current sheet. Click **Find next** to move through the matches.

5. Formulas

A formula starts with `=` and recalculates automatically whenever a cell it uses changes. For example, in the budget, cell E4 holds `=C4*D4`: quantity times unit cost.

Building blocks

You can use	Examples
Numbers, text in quotes, TRUE and FALSE	<code>=42</code> , <code>= "Paid"</code> , <code>=TRUE</code>
Cell references and ranges	<code>=B2</code> , <code>=SUM(B2:B9)</code> , <code>=SUM(A1:C3)</code>
Arithmetic	<code>+</code> <code>-</code> <code>*</code> <code>/</code> and brackets, as in <code>=(B2+B3)*1.2</code>
Comparisons	<code>=</code> <code><></code> <code><</code> <code><=</code> <code>></code> <code>>=</code> , as in <code>=B2>100</code>

Functions

Function	What it does	Example
<code>SUM</code>	Adds numbers. Text and empty cells are ignored.	<code>=SUM(E4:E9)</code>
<code>AVERAGE</code>	Average of the numbers in a range.	<code>=AVERAGE(D4:D9)</code>
<code>IF</code>	Chooses between two results. It always takes three parts: a test, the result when true, and the result when false.	<code>=IF(E4>1000,"Review","OK")</code>
<code>SHEET</code>	Reads a cell on another worksheet.	<code>=SHEET("Budget","E11")</code>
<code>LUA</code>	Runs a short Lua calculation.	<code>=LUA("return cell('E11') * 1.05")</code>

The quickest way to total a column is to select it and choose **Home, Editing, AutoSum, Sum**. Julretsu adds a `SUM` formula directly below your selection.

When a formula shows an error

Shows	Meaning
#DIV/0!	A division by zero, or an average of no numbers.
#VALUE!	Text was used where a number was needed.
#REF!	A reference points outside the sheet.
#NAME?	An unknown function name.
#CYCLE!	A formula depends on itself, directly or through other cells.
#NUM!	A result too large to represent.
#PARSE!	The formula could not be read. Check brackets and quotes.

Tip. Turn on **View, Show formulas** to see every formula in the grid instead of its result.

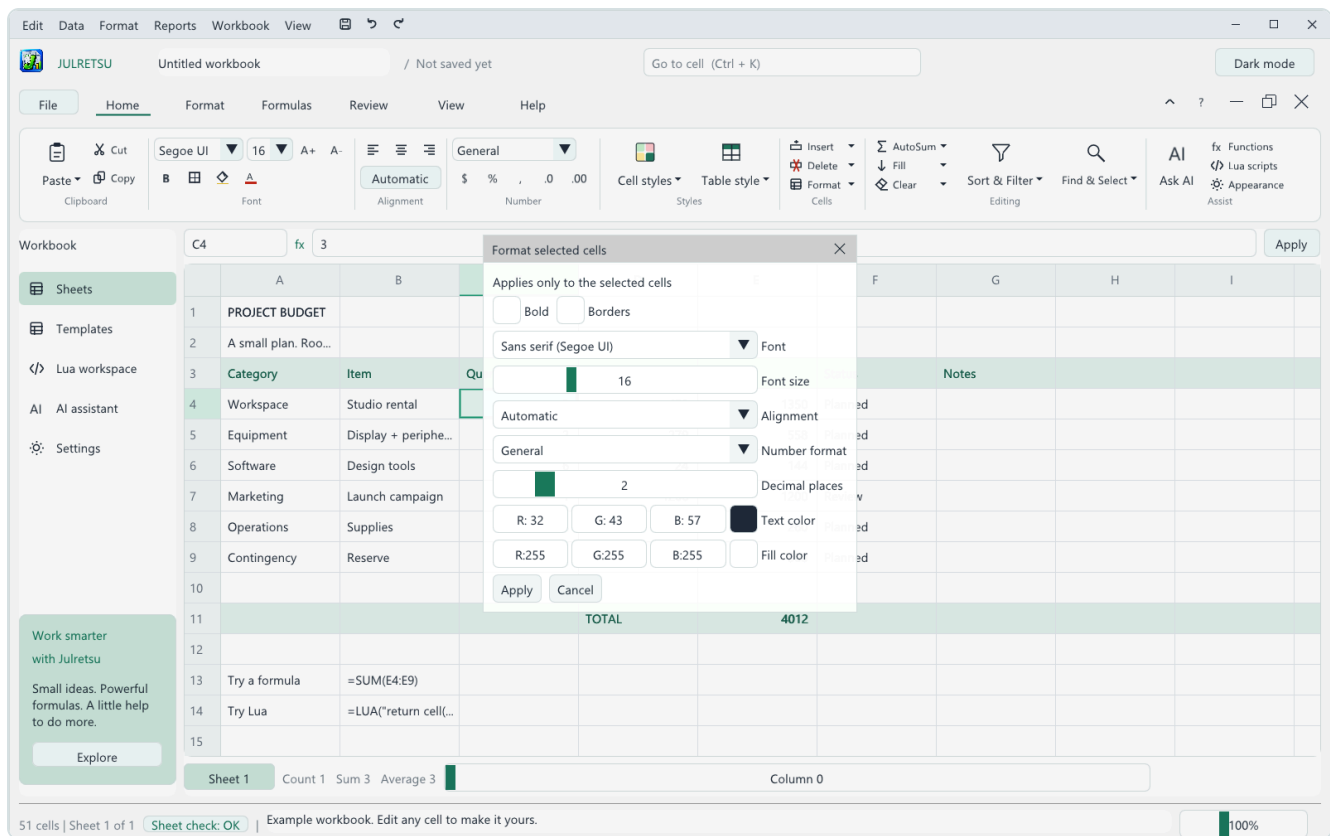
6. Formatting your sheet

The **Home** ribbon holds the everyday formatting tools. They apply to the selected cells. If you selected whole rows by clicking their row numbers, they format those rows instead.

Group	Tools
Clipboard	Paste (or paste values), Cut and Copy.
Font	Font, size, bigger and smaller text, bold, borders, fill colour and font colour.
Alignment	Left, centre, right, or Automatic (numbers right, text left).
Number	General, Number, Currency, Percentage, Date and Comma formats, plus fewer or more decimal places.
Styles	Ready made cell styles such as Good, Bad, Neutral, Title, Heading, Total, Input and Note, and table styles with a header row and banded rows.
Cells	Insert and delete rows, columns and worksheets, and open Format cells.
Editing	AutoSum, Fill, Clear, Sort & Filter, and Find & Select.

The Format cells dialog

For full control, choose **Format, Format selected cells** (or **Cells, Format, Format cells**). You can set the font family (sans serif, serif or monospace), size, bold, borders, alignment, number format, decimal places, and text and fill colours.



Format cells: every formatting option in one place

Number formats

Format	1234.5 appears as
General	1234.5
Number (2 decimals)	1234.50
Currency	\$1,234.50
Comma	1,234.50
Percentage (0.25 with 0 decimals)	25%
Date	Excel style date serial numbers shown as a date, for example 2026-09-20

Formatting whole rows

The **Format** tab has quick tools for entire rows: bold rows, a green fill, fewer or more decimals, and reset. They are handy for header and total rows.

Formatting a table in one click

Select your table including its header row, then choose **Home, Styles, Table style** and a colour. Julretsu makes the header bold and coloured, adds borders, and shades every other row.

7. Rows, columns, sorting and filtering

Inserting and deleting

Select a cell, then use **Home, Cells, Insert** or **Delete** (also on the **Data** menu). Rows are inserted above the selected cell and columns before it. Formulas elsewhere are updated so they still point at the right cells.

Note. Julretsu will not delete a row or column that other formulas depend on, because that would silently break them. It also asks you to remove `SHEET` and `LUA` references before inserting or deleting, since those cannot be adjusted automatically.

Sorting

1. Select the rows to sort, including every column that belongs to them. Leave the header row out.
2. Click a cell in the column to sort by, so it becomes the active cell.
3. Choose **Home, Editing, Sort & Filter**, then **Sort A to Z** or **Sort Z to A**.

Safety net. If columns beside your selection hold data in the same rows, Julretsu warns you straight away, because sorting only part of a table mixes up rows. Choose **Undo change** and select the whole table.

Filtering

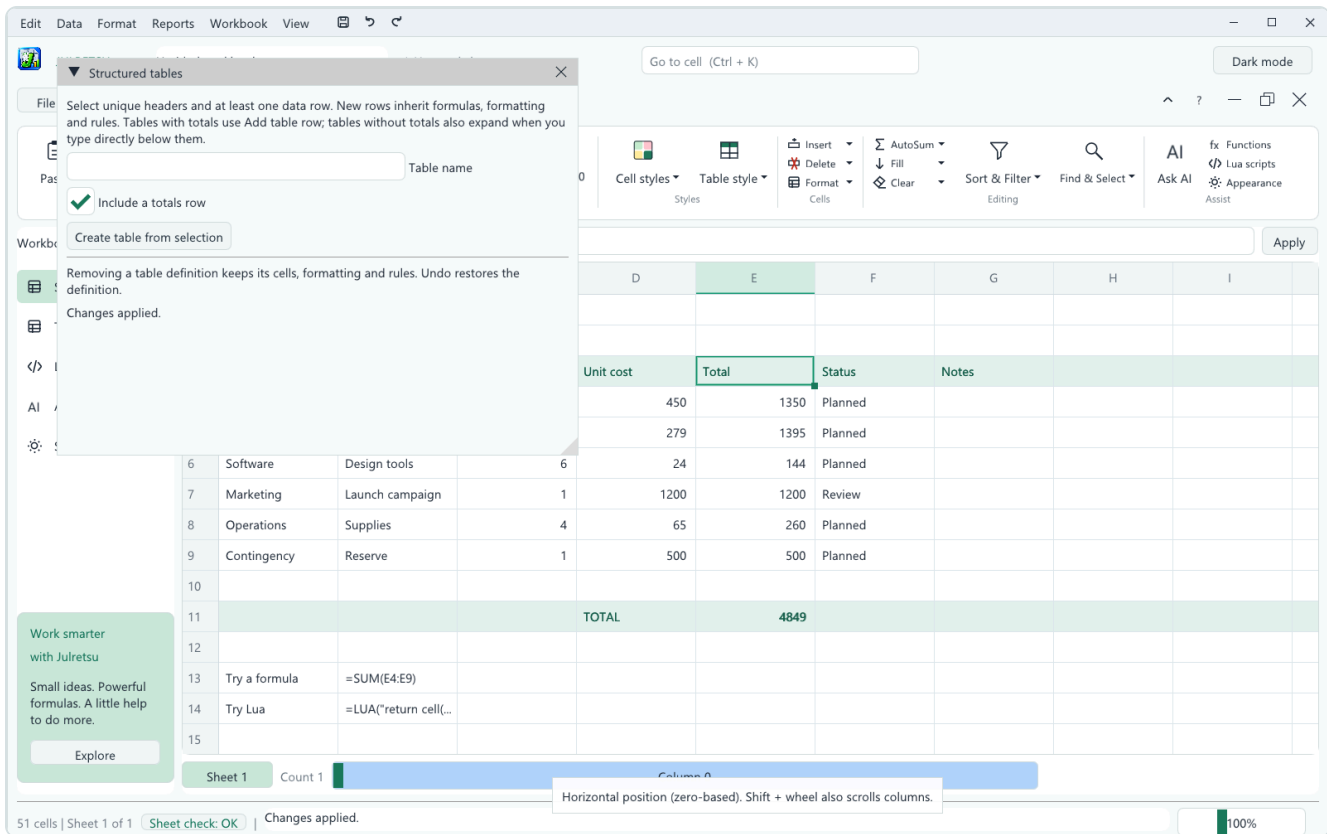
Click a cell in the column you want to filter, then choose **Data, Filter active column**. Type the text to look for. Only rows containing that text stay visible (capital letters must match), and row 1 always stays visible as your header. Choose **Data, Clear filter** to show everything again. Filtering never deletes data. To protect hidden rows, clear the filter before sorting, pasting or deleting.

Freezing headings

On the **View** menu choose **Freeze first row**, **Freeze first column**, or both, so your headings stay in view while you scroll.

Structured tables

A structured table keeps a block of data together, so that it grows correctly. Select your header row and at least one data row, then choose **Data, Structured tables**, give the table a name and click **Create table from selection**. Tick **Include a totals row** to have Julretsu add and maintain a total under each numeric column.



Turning a selection into a structured table

- New rows inherit the formulas, formatting and validation rules of the row above, so a price or total column keeps calculating.
- A table with a totals row grows with **Add table row**, which inserts the new row above the total and extends the total to cover it.
- A table without a totals row also grows on its own when you type in the first empty row directly below it.
- **Remove table definition** stops the automatic behaviour and keeps every cell, format and rule as it is. Undo restores the definition.

Note. Headers must be filled in and different from each other, tables cannot overlap, and one table covers at most 100,000 cells. Remove a table definition before inserting or deleting rows and columns, and use Add table row to extend it instead.

8. Working with several worksheets

A workbook can hold up to 64 worksheets. Click the sheet button at the bottom left of the grid to switch sheets, or choose **Workbook, Manage sheets** to add, duplicate, rename, reorder or delete them.

To use a value from another sheet, use `SHEET` with the sheet name and cell address in quotes, for example `=SHEET("Q1 Sales","B12")`. If you rename a sheet, update the name inside your formulas as well.

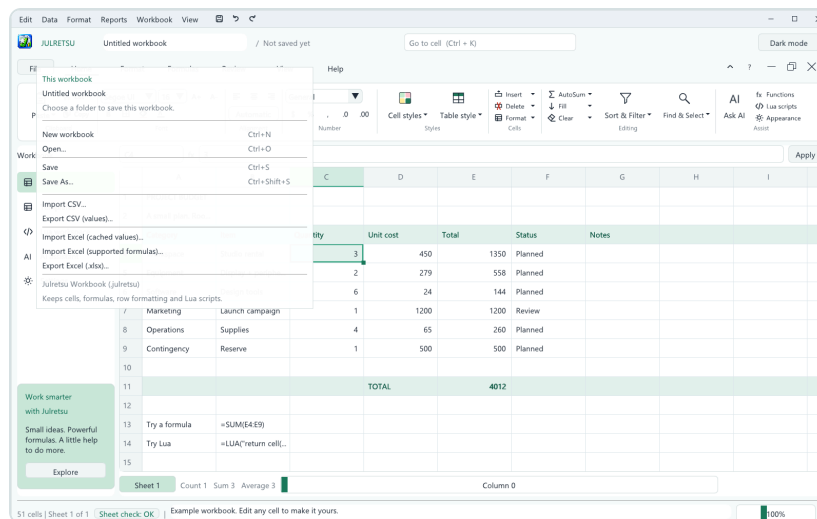
Careful. Deleting a worksheet removes all of its contents and cannot be undone. Julretsu asks you to confirm first.

9. Saving, opening and recovery

Julretsu workbooks

Save with **Ctrl** + **S**, the quick access Save button, or **File, Save**. The first time, choose a folder and a name. Use **File, Save As** (**Ctrl** + **Shift** + **S**) to save a separate copy. Open workbooks with **Ctrl** + **O**, or double-click a `.julretsu` file.

A `.julretsu` file keeps everything: values, formulas, formatting, all worksheets, your Lua script and the change history. Julretsu always asks before closing or replacing a workbook with unsaved changes.



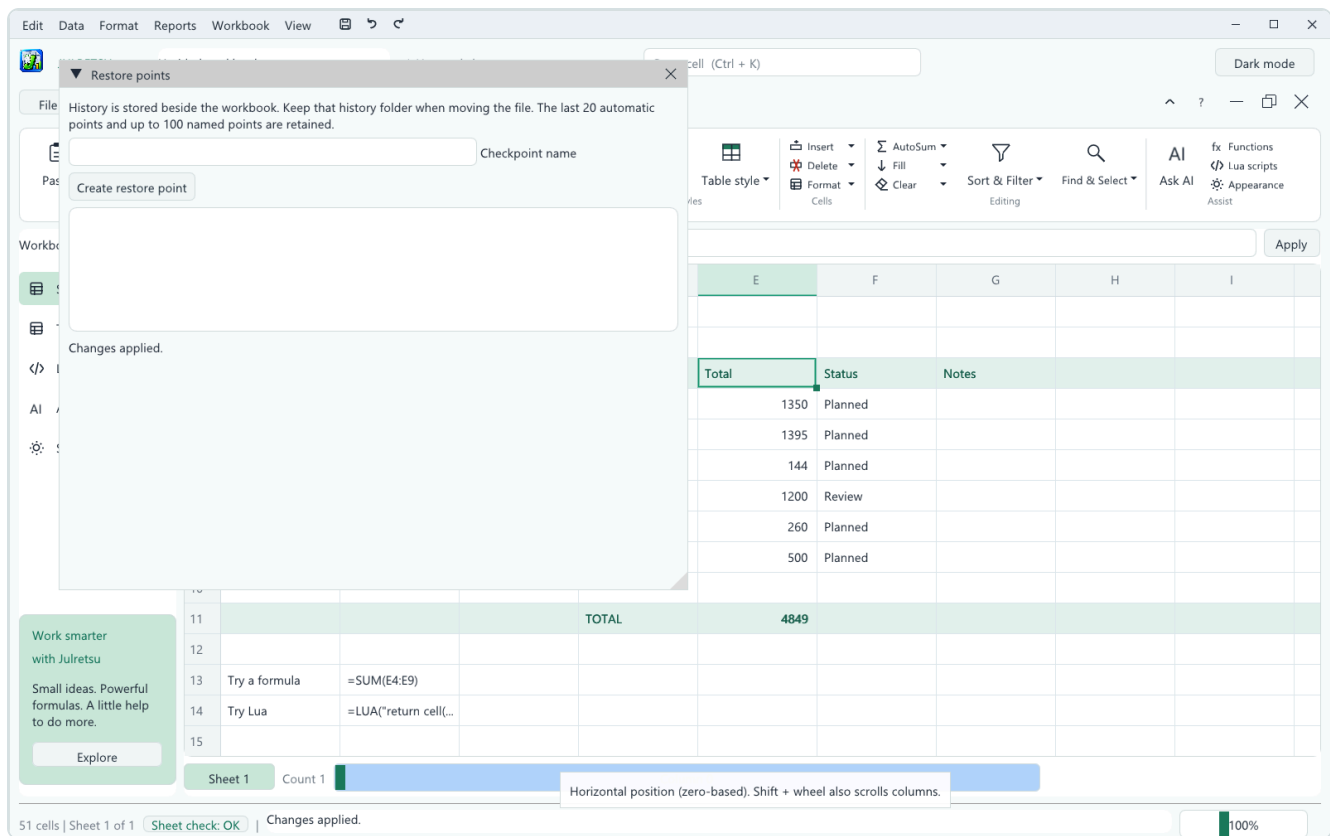
The File menu

Automatic recovery

After a minute of unsaved work, Julretsu quietly keeps a recovery copy and refreshes it every minute. If Julretsu or your computer closes unexpectedly, the next time you start Julretsu it offers to restore that copy. Choose **Restore**, then save the workbook with **Save As**. The recovery copy is removed whenever you save or close Julretsu normally. You can turn it off, or open it manually, from the **Workbook** menu.

Restore points

A restore point is a snapshot of the whole workbook that you can go back to later. Choose **Edit, Restore points**, or the **Restore points** button on the **Review** tab. Save the workbook first, then type a name and click **Create restore point**.



Restore points, with a preview of what changed

1. Select a point in the list to see what changed since then: how many cells differ, and which worksheets, formats or rules changed.
2. Click **Restore selected version**. Julretsu first saves a safety copy of your current work, called **Before restore**.
3. Check the result, then save to keep it. If it was not what you wanted, restore the safety copy instead.

Julretsu also takes a point named **Before save** each time you save, so a bad afternoon of edits is never the only copy you have.

Note. Points are kept in a history folder beside the workbook file, so move that folder with the file. Julretsu keeps the last 20 automatic points and up to 100 named ones.

10. CSV and Excel files

CSV files

Use **File, Import CSV** and **File, Export CSV**. Julretsu reads and writes UTF-8 CSV files, including values that contain commas, quotes or several lines. When importing, numbers and TRUE or FALSE are recognised and everything else becomes text. Formulas in a CSV file are never run.

Exported CSV files contain calculated values. Text that starts with `=`, `+`, `-` or `@` is protected with a leading apostrophe, so another spreadsheet program will not run it as a formula.

Excel workbooks

File, Import Excel reads every worksheet of an `.xlsx` file, with each cell's own formatting and any validation rules it can support. Choose **cached values** to bring in the results Excel last calculated, or **supported formulas** to keep arithmetic, comparisons, `SUM`, `AVERAGE` and `IF` as live formulas. References to another sheet arrive as `SHEET` formulas. Before anything loads, Julretsu shows exactly what will and will not come across, sheet by sheet.

File, Export Excel saves the whole workbook as an `.xlsx` file: every worksheet, its supported formulas, cell formatting, and dropdown, number and date rules that Excel understands. Formulas Julretsu cannot express in Excel are written as their calculated values, and the message afterwards says how many.

Note. Excel has its own limits, so sheet names must be unique and at most 31 characters, dropdown lists must fit 255 characters without commas or quotes, and validation ranges must not overlap. Julretsu tells you which rule to change rather than dropping it quietly.

Note. Excel features such as merged cells, charts, column widths and comments are not imported or exported, and workbooks using the 1904 date system are not supported. Keep a `.julretsu` copy of your work for full fidelity.

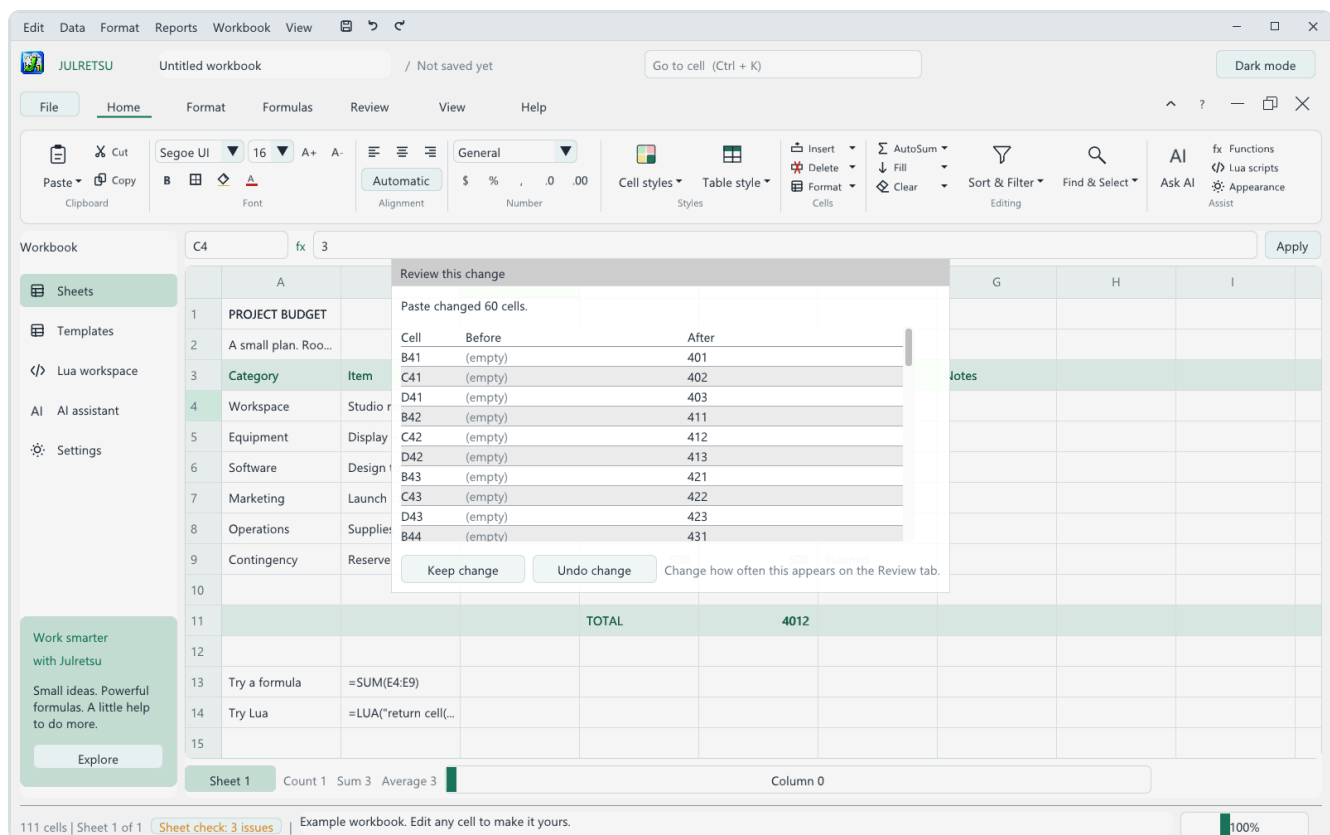
11. The safety net

Julretsu looks out for the everyday mistakes behind costly spreadsheet errors. It never blocks you; it explains what looks wrong and lets you decide.

Review before keeping risky changes

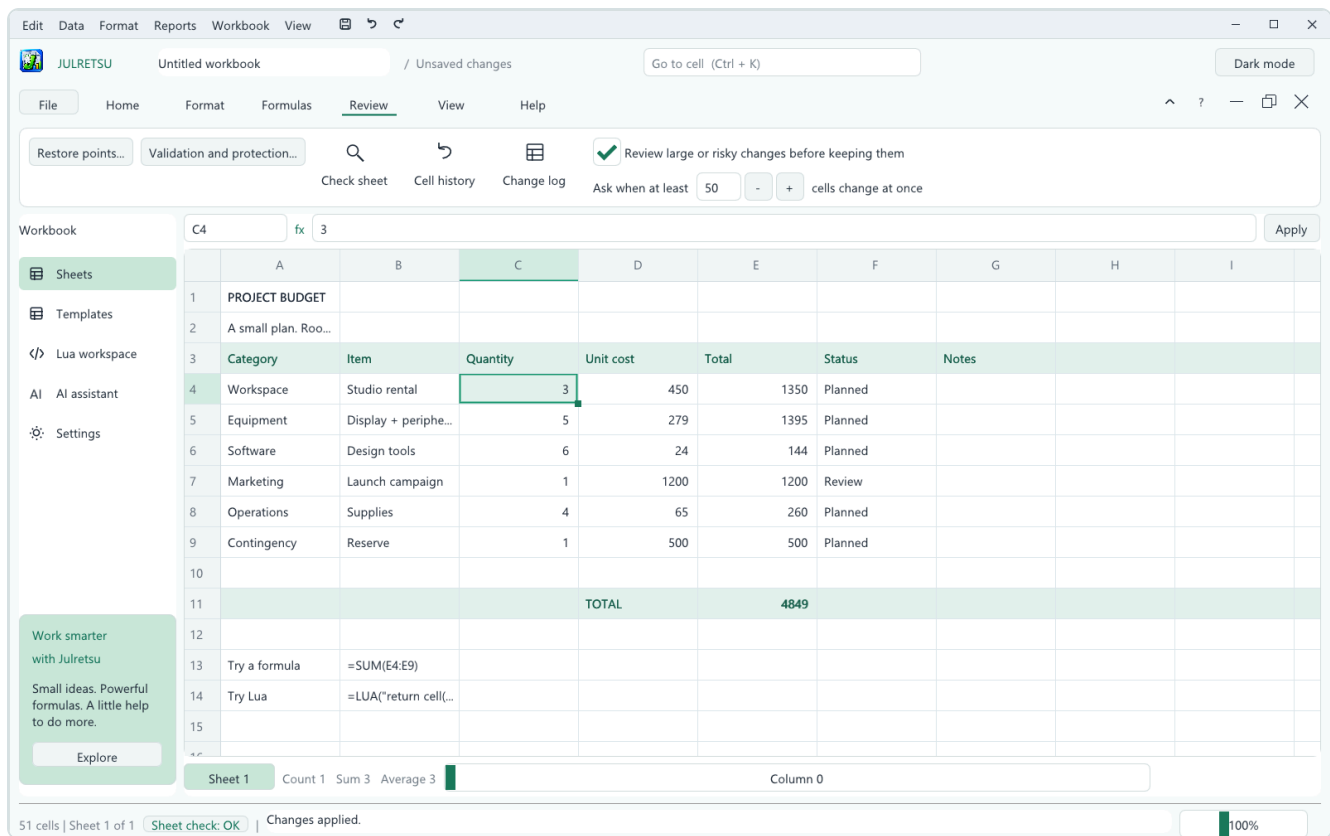
After a large paste, fill, clear or macro, after deleting a row or column that held data, after a sort that left neighbouring columns behind, or when formulas are overwritten, Julretsu shows **Review this change**. The change is already visible in your sheet and the dialog lists every affected cell with its old and new value.

- **Keep change** (Enter) accepts it.
- **Undo change** (Esc) puts everything back exactly as it was.



Review this change after a large paste

On the **Review** tab you can choose how many cells a change must touch before Julretsu asks, or turn reviews off. Partial sorts, deleted data, replaced formulas and out of scale numbers are always reviewed.



The Review tab

Instant typo check

If you type a number far larger or smaller than the rest of its column, for example 4500000 in a column of amounts around 450, the status bar warns you right away. Press **Ctrl + Z** if it was a slip.

Sheet check

The **Sheet check** badge in the status bar always shows whether your sheet has problems. Click it, or choose **Review, Check sheet**, to open the panel. Julretsu looks for:

- a typed number, or a different formula, inside a column of matching formulas;
- **SUM** or **AVERAGE** ranges that stop short of neighbouring numbers;
- numbers wildly out of scale with the rest of their column;
- numbers stored as text, which totals silently skip;
- blank rows that split a table and break sorting and filtering.

Each problem has **Go to**, a one click fix when a safe fix exists, and **Ignore**. Red dots mark problems likely to give wrong results. Every fix can be undone.

Julretsu - Untitled workbook / Not saved yet

Go to cell (Ctrl + K)

Dark mode

File Home Format Formulas Review View Help

Clipboard: Paste, Cut, Copy

Font: Segoe UI, 16, A+, A-, Automatic, Alignment, Number

Styles: Cell styles, Table style

Cells: Insert, Delete, Format, AutoSum, Fill, Clear, Sort & Filter, Find & Select

AI: Ask AI, Functions, Lua scripts, Appearance

Workbook: C4, fx, 3

Sheets: Sheets, Templates, Lua workspace, AI assistant, Settings

Work smarter with Julretsu: Small ideas. Powerful formulas. A little help to do more. Explore

	A	B	C	D	E	F	
1	PROJECT BUDGET						
2	A small plan. Roo...						
3	Category	Item	Quantity	Unit cost	Total	Status	Notes
4	Workspace	Studio rental	3	450	1350	Planned	
5	Equipment	Display + periphe...	2	279	558	Planned	
6	Software	Design tools	6	24	999	Planned	
7	Marketing	Launch campaign	1	1200	1200	Review	
8	Operations	Supplies	4	65	260	Planned	
9	Contingency	Reserve	1	500	500	Planned	
10							
11			TOTAL		4367		
12							
13	Try a formula	=SUM(E4:E9)					
14	Try Lua	=LUA("return cell(...					
15							

Sheet 1 Count 1 Sum 3 Average 3 Column 0

51 cells | Sheet 1 of 1 Sheet check: 2 issues Example workbook. Edit any cell to make it yours. 100%

SHEET CHECK 2 to review (2 may give wrong results)

Checks run as you edit: formula patterns, totals that skip rows, out-of-scale numbers, numbers stored as text and blank rows inside tables.

- E6 is a typed number inside a column of formulas

The cells above and below are calculated, but E6 holds a fixed 999. It will not update when its inputs change.

Go to Use =C6*D6 Ignore

- E11's SUM leaves out 1 row

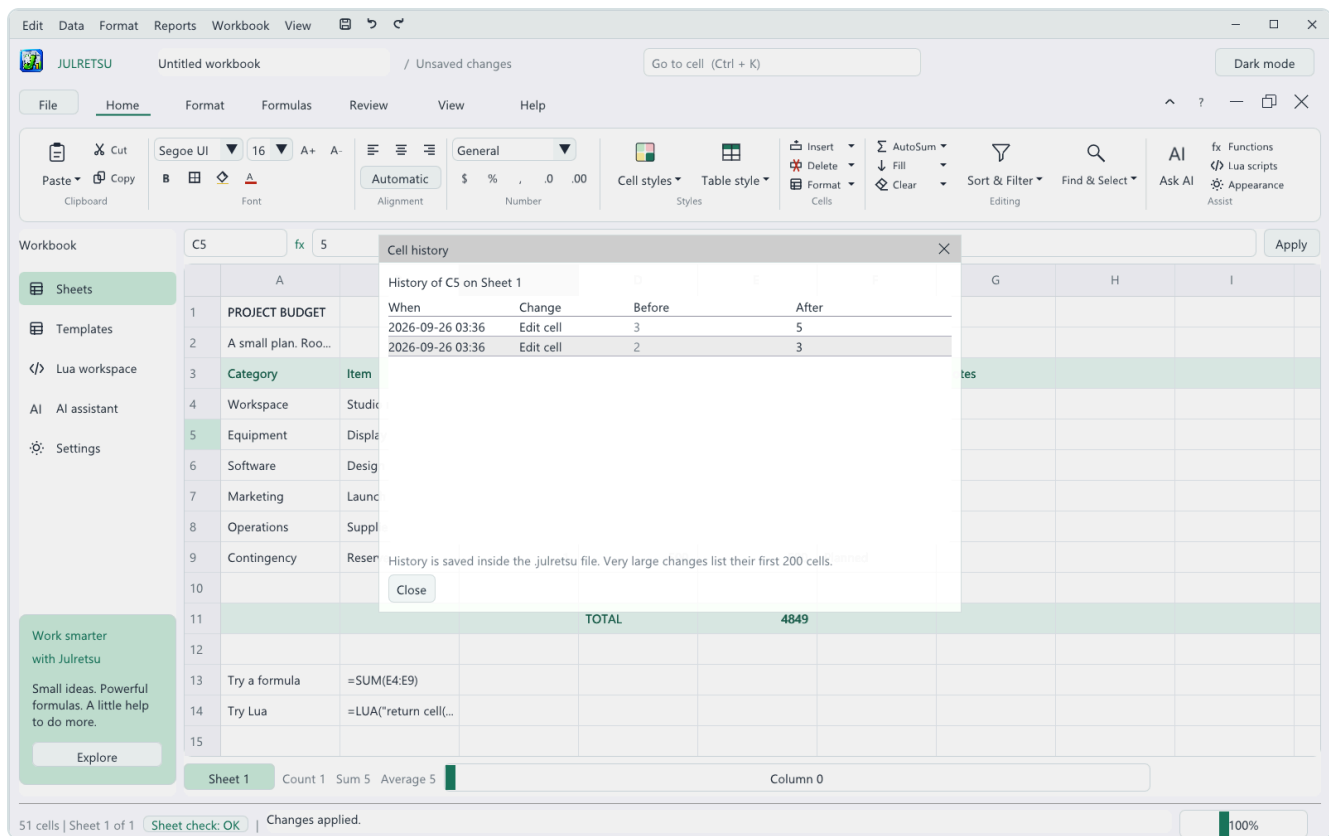
It covers E4:E8, but E9 is a number right next to that range and is not included.

Go to Extend to E4:E9 Ignore

Sheet check found a typed value in the Total column and a total that skips a row

Cell history and the change log

Right-click a cell and choose **Cell history**, or use **Review, Cell history**, to see every change to that cell: when it happened, what caused it, and the value before and after. **Review, Change log** lists every change on the sheet.

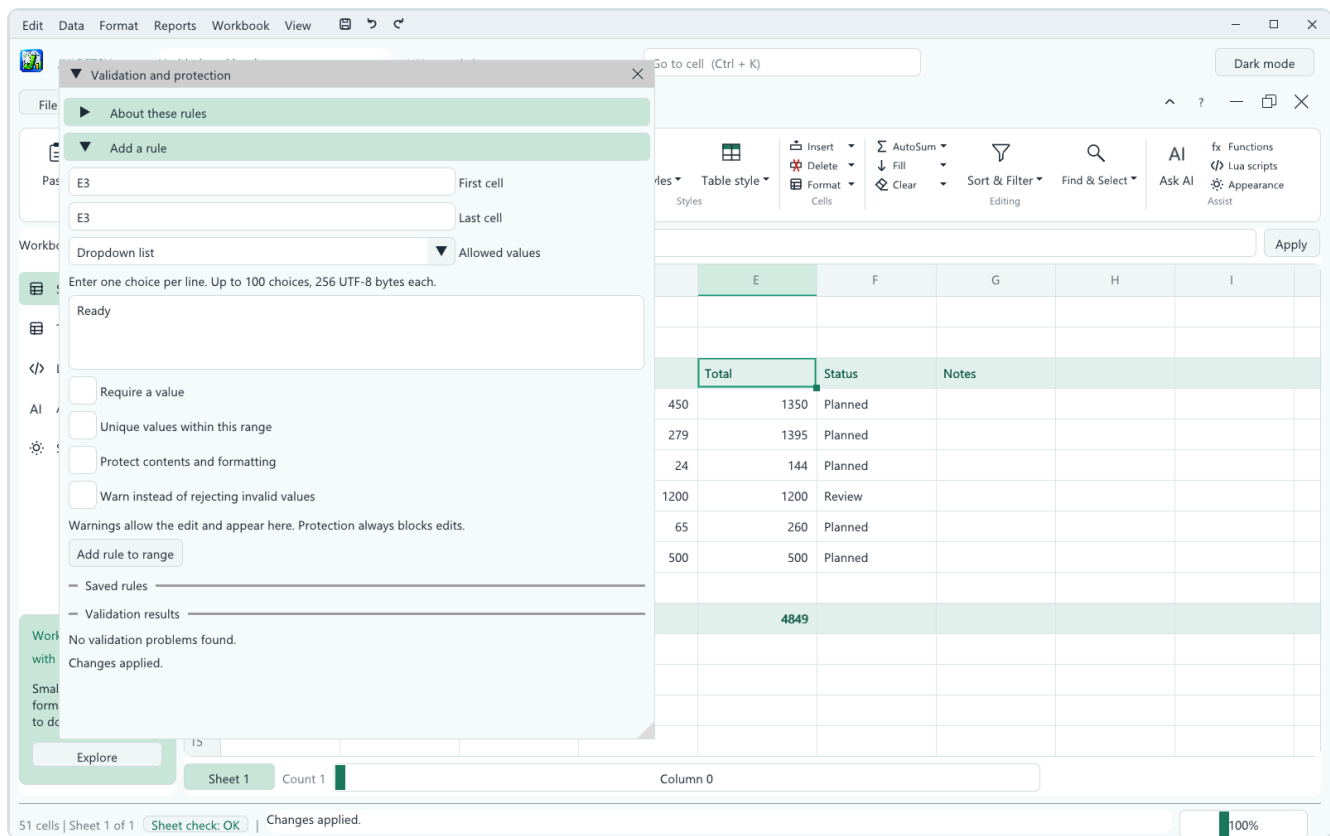


The history of a single cell

History is saved inside the `.julretsu` file, so it travels with the workbook. Julretsu keeps the latest 1,000 changes, and very large changes list their first 200 cells. If you share a file and prefer not to include its history, open the change log and choose **Clear history**.

Validation rules and protection

A validation rule says what a cell is allowed to hold, so a wrong value is refused as it is typed. Select the range, then choose **Data, Validation and protection**, or the button on the **Review** tab.



Adding a dropdown list to a column

Allowed values	What it accepts
Any value	Anything, useful together with Require a value or protection.
Number, Whole number	Values between a lowest and highest amount, inclusive.
Date	Dates written as YYYY-MM-DD, between two dates.
Dropdown list	One of the choices you type, one per line, up to 100 of them.

- **Require a value** refuses an empty cell, and **Unique values within this range** refuses a duplicate.
- **Protect contents and formatting** blocks edits to the range. Protected formulas still recalculate.
- **Warn instead of rejecting invalid values** lets the edit through and lists it under **Validation results**.

Rules apply to everything, not just typing: a paste, a fill, a sort, a macro or an AI proposal that would break a rule is refused as a whole, so you never end up with half of it applied. Adding a rule to a sheet that already has problems keeps your data and lists the problems for you to fix. Click a listed address to jump to that cell.

Careful. Protection prevents accidents; it is not a password or a permissions system. Anyone who has the workbook can remove its rules. Rules are saved only in `.julretsu` files, not in CSV or Excel exports, and a workbook with rules cannot be opened by Julretsu 1.1.0 or earlier.

12. Charts, reports and PDF

1. Select the cells you want in your report. If only one cell is selected, Julretsu uses every filled cell on the sheet.
2. Choose **Reports, Chart / print selected range**.
3. Give the report a title and pick the **Chart column** that holds the numbers to chart. Choose a bar or line chart, portrait or landscape, and whether to repeat the first row on every page.
4. Click **Export chart + table (HTML)** and choose where to save the file. Open it in any web browser to see the chart and table together, then use the browser's **Print** command to print it or save it as a PDF. This works on Windows, Mac and Linux.
5. On Windows you can also click **Print / Save as PDF** to print directly. To make a PDF, choose **Microsoft Print to PDF** as the printer.

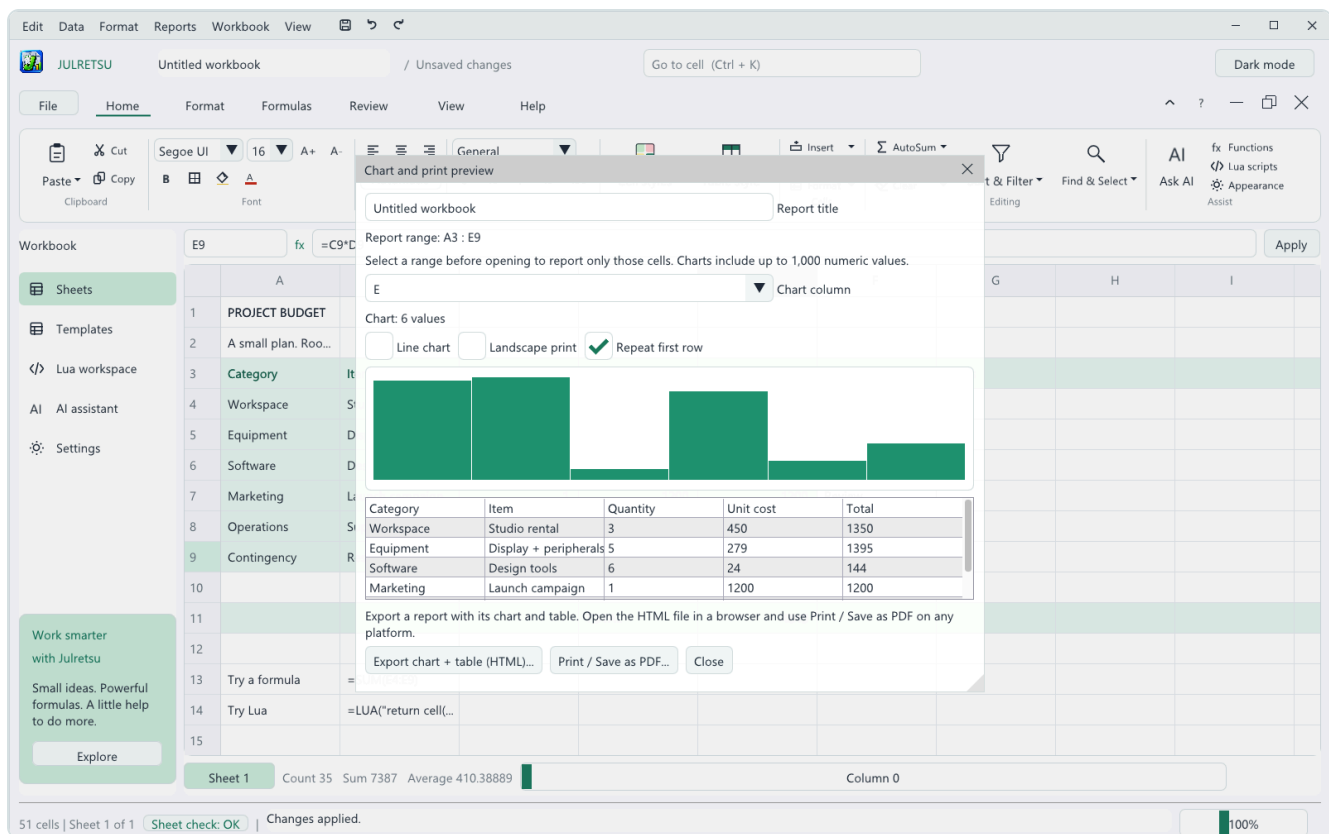


Chart and print preview

Note. A report can include up to 12 columns and 10,000 cells, and the chart uses up to 1,000 numbers from the chart column. A single number is shown as one bar so it stays visible. Reports are snapshots and are not saved in the workbook. The exported HTML file holds everything it needs, so you can email it or open it on another computer.

Sample templates

Click **Templates** in the navigation bar to open a ready-made workbook filled with fictional sample data and working formulas. Choose one from the **Sample workbook** list and click **Use selected template**. Julretsu offers to save your current workbook first.

Template	What it shows
Project budget	Planned and actual costs by category, with totals and the difference.
Monthly sales	Online and store sales for twelve months compared with a target. A good line chart.
Household expenses	Planned and actual spending by category, with the amount remaining.
Inventory planning	Stock on hand, reorder levels, unit costs and the value of each product.
Weekly project hours	Design, development and testing hours for twelve weeks, with weekly totals.

Templates are a quick way to try charts: open one, select its table and choose **Reports, Chart / print selected range**.

13. Lua macros

Lua macros automate repetitive edits. Open the **Lua workspace** from the navigation bar or **Home, Assist, Lua scripts**. The editor has a **Code** tab for your script and a **Quick reference** tab listing what a macro can call. Below the code, a line shows where the cursor is, and **Output** reports what the last run did.

- `cell('A1')` reads a cell as it was before the macro started.
- `set('A1', value)` queues a change.
- Click **Run macro**, or press `Ctrl1 + Enter` while writing. All changes are applied together, and **Undo macro** reverts them in one step.

This example adds 1 to every quantity in the budget:

```
for row = 4, 9 do set('C' .. row, cell('C' .. row) + 1) end
```

The screenshot shows the Julretsu application interface. The main window displays a spreadsheet titled 'PROJECT BUDGET' with columns A through E. The data includes categories like Workspace, Equipment, Software, Marketing, Operations, and Contingency, along with their quantities, unit costs, and totals. The 'Quantity' column (C) is highlighted, and the value '5' is visible in row 5. The 'TOTAL' row (11) shows a total of 4849.

On the right side, the 'Lua workspace' editor is open. It has a 'Code' tab and a 'Quick reference' tab. The 'Code' tab contains the following Lua script:

```
-- Macros commit once. Undo restores every edit.
-- cell() reads the sheet before the macro.
for row = 4, 9 do
  local quantity = cell('C' .. row)
  set('C' .. row, quantity + 1)
end
```

Below the code, there is a 'Run macro' button and an 'Undo macro' button. The 'Output' section shows the result of the last run: 'Run a macro to see its result here.'

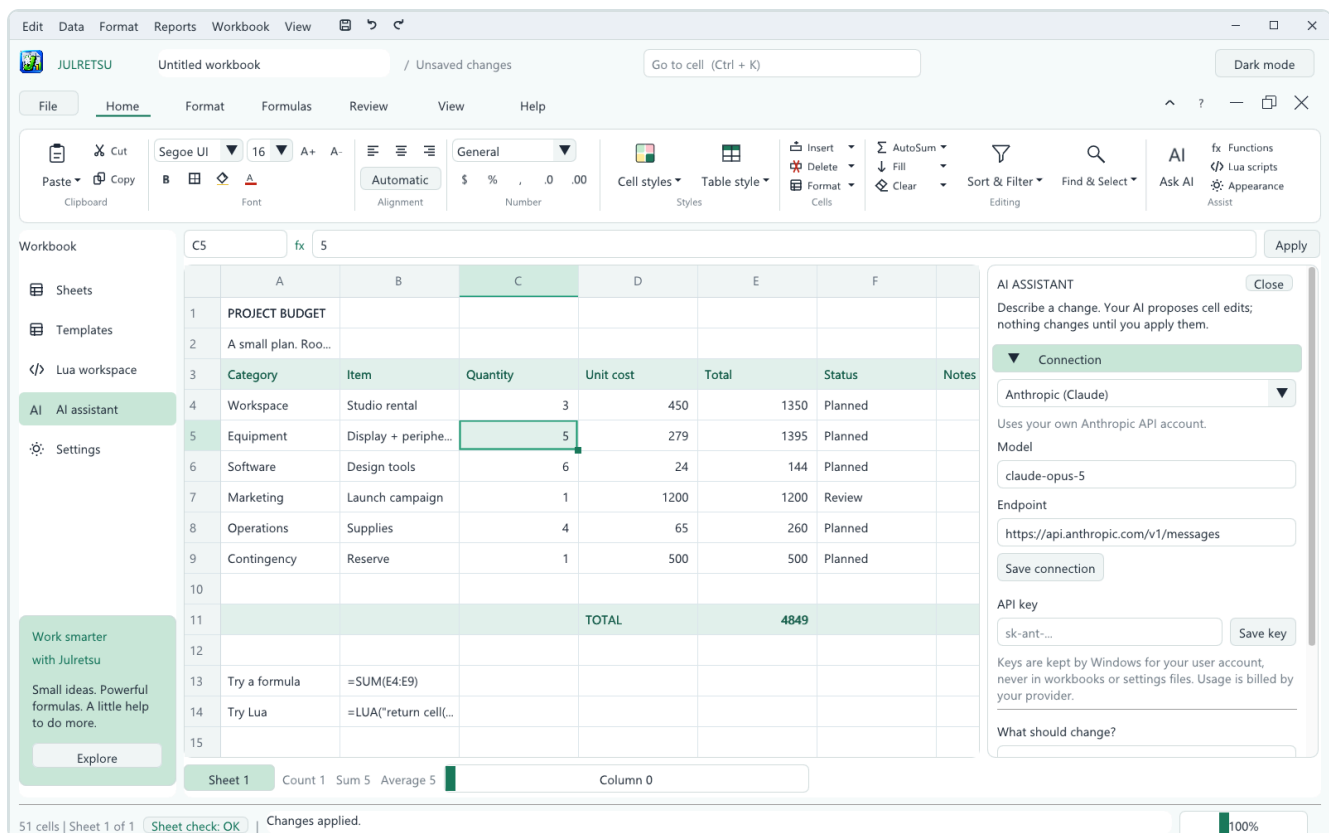
The Lua workspace

Note. Macros run in a safe sandbox with no access to your files, the network or the rest of your computer, and with limits on time and memory. A macro that fails leaves your sheet unchanged. The script is saved with the workbook but never runs by itself.

14. The AI assistant (optional)

Julretsu works fully offline. If you want, you can connect your own AI provider and ask for changes in plain language, such as "add a 10% contingency row under the total".

1. Open **AI assistant** from the navigation bar or **Home, Assist, Ask AI**.
2. Under **Connection**, choose **Anthropic (Claude)** or an **OpenAI compatible** server such as OpenAI, Ollama or LM Studio, enter the model name, and save your API key.
3. Describe the change and click **Propose edits**.
4. Review each proposed edit with its before and after values, untick anything you do not want, and click **Apply**.



The AI assistant panel

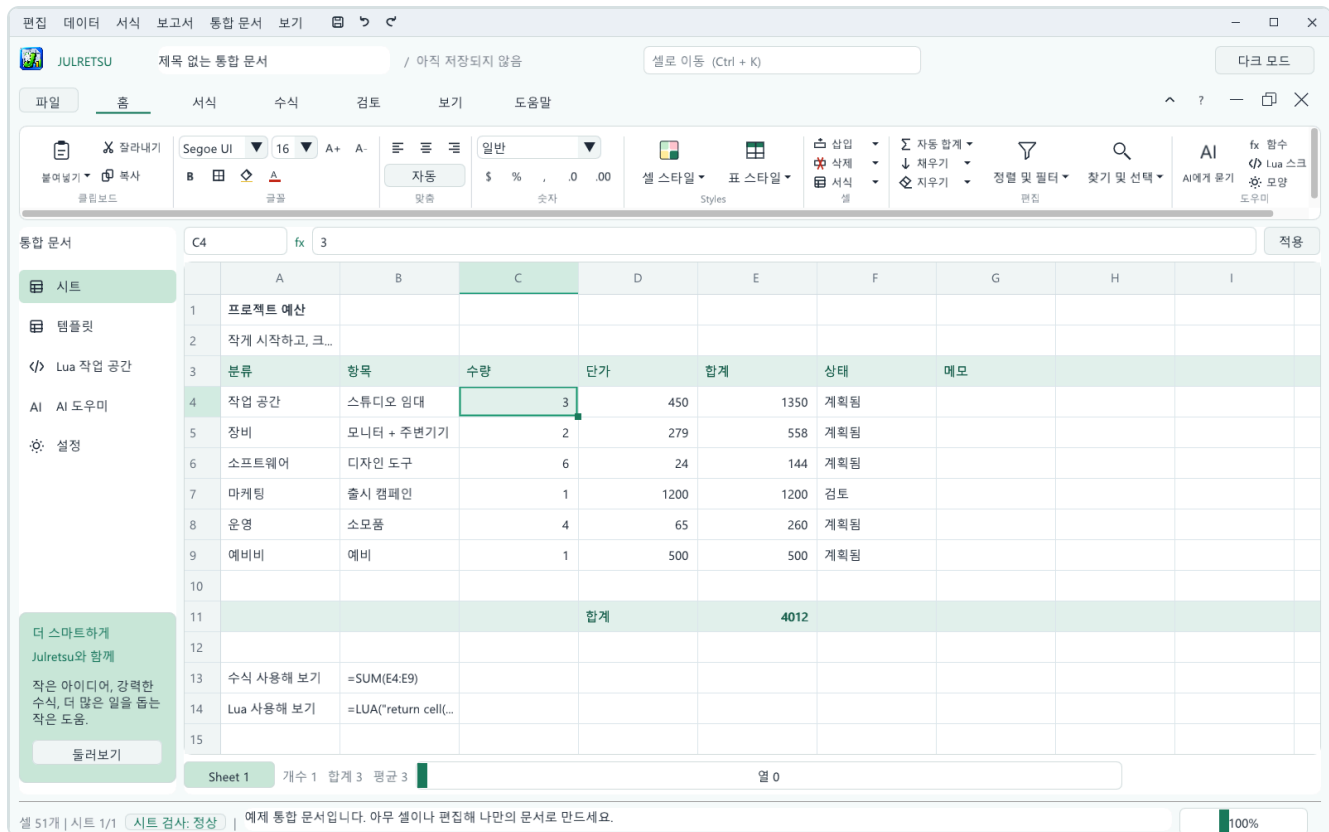
- Your API key is stored by Windows Credential Manager for your user account, never in workbooks or settings files.
- The panel tells you how many cells will be sent and to which server before you send anything. At most 2,000 filled cells are sent.
- Edits that contain Lua formulas start unticked. Nothing changes until you click Apply, and one **Ctrl + Z** undoes the whole proposal.
- Usage is billed by your provider under your own account.

Note. AI connections are currently available in the Windows version of Julretsu.

15. Language, views and themes

Choosing a language

Julretsu speaks English, Korean and Japanese. The first time you start it, it follows the language of your computer. To change it, click **Settings** in the navigation bar and pick a language under **Language**, or use **View, Language**. Your choice is remembered on this device.



The same workbook with the interface set to Korean

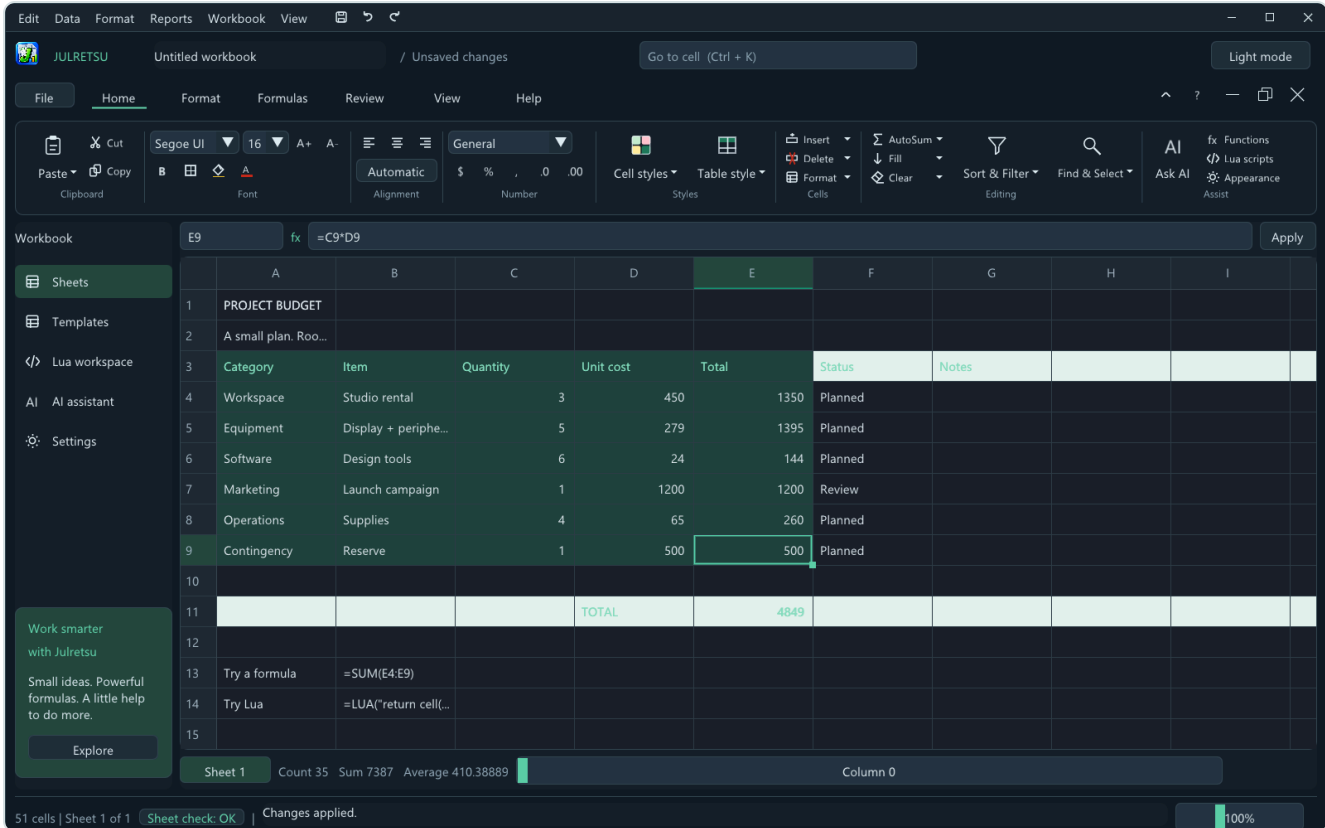
The language applies to menus, buttons, messages, the sheet check, exported reports and new sample workbooks. Your own cell contents are never translated, and neither are workbooks you already saved.

You can type Korean and Japanese into any cell in any language setting, using your usual input method. Julretsu loads the characters it needs from the fonts installed on your computer, so text stays sharp at every zoom level.

Note. On Linux, install a CJK font such as Noto Sans CJK (on Ubuntu or Debian: `sudo apt install fonts-noto-cjk`) so Korean and Japanese characters can be drawn. Windows and macOS already include them.

Dark mode

Click **Dark mode** at the top right, or use **View, Dark appearance**. Julretsu remembers your choice.



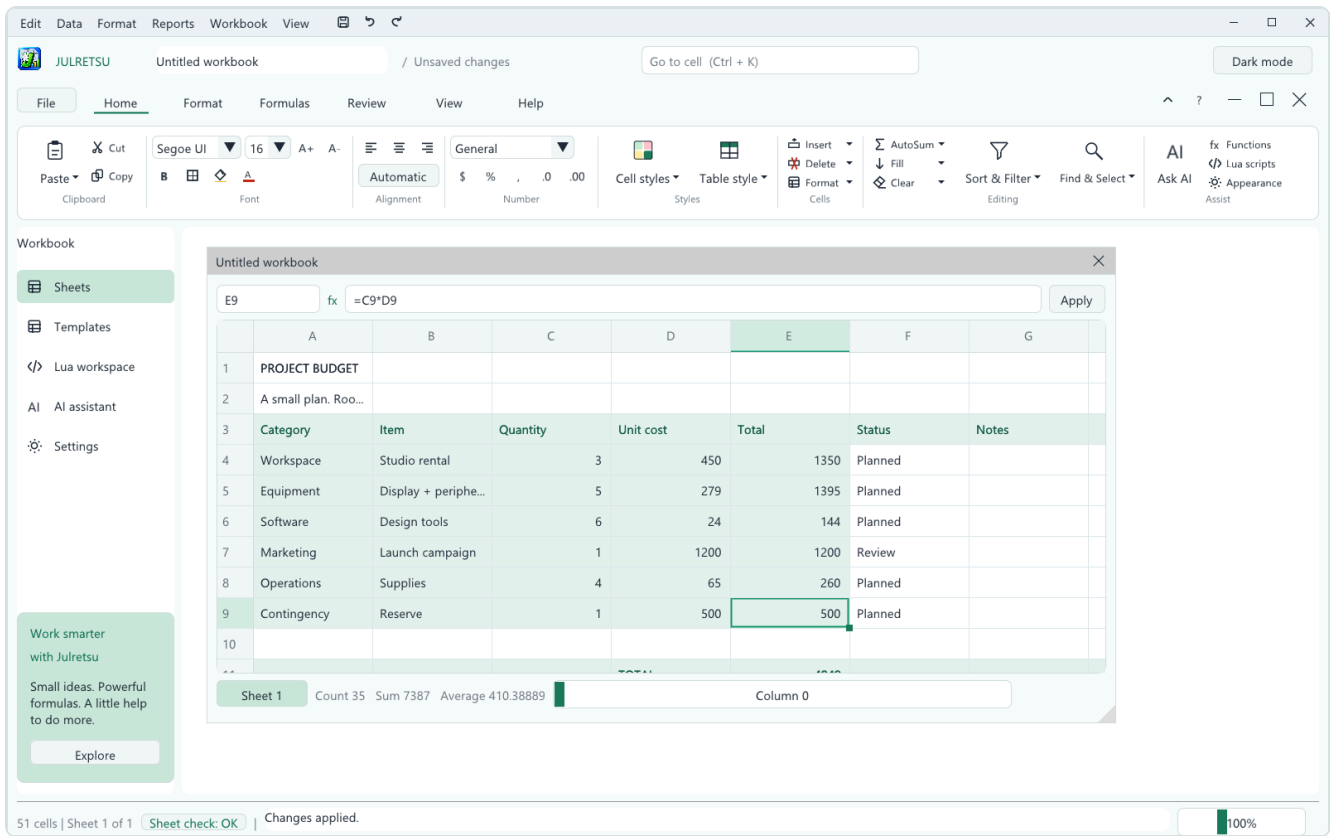
Julretsu in dark mode

Other view options

- **Zoom:** drag the slider at the bottom right, from 75 to 150 percent.
- **Gridlines** and **Show formulas:** on the View menu and the View tab.
- **Long text spills into empty cells:** lets long text run across empty neighbouring cells, as older spreadsheets do.
- **Collapse the ribbon:** click the arrow beside the ribbon tabs for more room. Click any tab to bring it back.

The workbook window buttons

The buttons at the right of the ribbon tabs control the workbook itself. **Minimize** shrinks it to a small bar at the bottom of the window, **Restore** places it in a window you can move and resize inside Julretsu, and **Close** closes it (you are asked to save first) and starts a blank workbook.



A restored workbook window inside Julretsu

16. Keyboard shortcuts

On a Mac, use **Cmd** wherever this table says **Ctrl**.

Action	Keys
New, open, save, save as	Ctrl + N , Ctrl + O , Ctrl + S , Ctrl + Shift + S
Undo, redo	Ctrl + Z , Ctrl + Y
Copy, cut, paste, paste values	Ctrl + C , Ctrl + X , Ctrl + V , Ctrl + Shift + V
Edit the selected cell	F2 or double-click
Keep or cancel an edit	Enter , Esc
Clear contents	Delete
Move the active cell	Arrow keys, Tab , Page Up , Page Down
First cell, last cell	Ctrl + Home , Ctrl + End
Go to a cell	Ctrl + K
Find	Ctrl + F
Select a range	Shift + click
Add or remove rows from a selection	Ctrl + click row numbers
Scroll rows, scroll columns	Mouse wheel, Shift + mouse wheel
Keep or undo a reviewed change	Enter , Esc

17. Troubleshooting

Problem	What to do
Windows says "Windows protected your PC" when installing.	Click More info , then Run anyway . See chapter 2.
macOS says it cannot check Julretsu for malicious software.	Control-click Julretsu in Applications and choose Open , or use Open Anyway in Privacy & Security.
Linux shows a message about zenity or kdialog.	Install one: <code>sudo apt install zenity</code> . Julretsu uses it for open and save dialogs.
Julretsu will not start and mentions OpenGL or graphics.	Update your graphics driver. Julretsu needs OpenGL 3.3 on Windows and Linux.
A cell shows ###.	The number is too wide for the cell. Zoom in, or read it in the formula bar.
A formula shows an error code.	See the error table in chapter 5, and check the Sheet check panel.
I cannot sort, paste or delete.	Clear any active filter first (Data, Clear filter).
Inserting a row is refused.	Remove <code>SHEET</code> or <code>LUA</code> references first; they cannot be adjusted automatically.
The recovery prompt appears when I start Julretsu.	Julretsu closed unexpectedly last time. Choose Restore to get your work back, or Keep current workbook to discard the copy.
Double-clicking a <code>.julretsu</code> file does not open Julretsu.	On Windows, run the installer again and tick Open .julretsu workbooks with Julretsu . On a Mac, open files with File, Open .

18. Uninstalling Julretsu

- **Windows:** open **Settings, Apps, Installed apps**, find Julretsu, and choose **Uninstall**. You can also use **Uninstall Julretsu** in the install folder.
- **Mac:** drag Julretsu from Applications to the Trash.
- **Linux (.deb):** run `sudo apt remove julretsu`.
- **Linux (.tar.gz):** delete the extracted folder.

Uninstalling never deletes your workbooks. Your appearance preference is stored in your user settings folder and can be deleted by hand if you wish.

19. License and contact

Julretsu is free to use under the Julretsu Software License. You may install and use it for personal, educational and business work, and the workbooks you create belong to you. Julretsu itself remains the property of Matthew Menchinton and Circuitspecter Studio. It is not open source, and you may not sell, redistribute or modify it. The full license is shown in the installer and included as the `LICENSE` file.

Julretsu includes third party components, each under its own license, listed in `THIRD_PARTY_NOTICES.md`: Dear ImGui, GLFW, Lua, sol2, stb, miniz and pugixml.

For questions or permission requests, contact the owner through [**github.com/Deathbringer98**](https://github.com/Deathbringer98).

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